



MULTI-AGENT STOCHASTIC GAMES FOR AUTONOMOUS INTERSECTIONS

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THE PROBLEM

- **Conventional Traffic Management:** Traditional traffic lights rely on static, inefficient time-allocation models.
- **Socioeconomic Toll:** Millions of hours wasted in congestion, inflating carbon emissions and fuel waste.
- **The Scalability Barrier:** Human drivers lack real-time cooperative computation capabilities, leading to systemic bottlenecks at signalized urban junctions.



SOLUTION

- **Autonomous Intersection Management (AIM):** Replacing physical traffic signals with decentralized algorithmic protocols.
- **The Junction as a Grid:** The intersection is mapped as a discrete conflict zone regulated by multi-agent coordination.
- **Connected Rationality:** High-speed Vehicle-to-Infrastructure (V2I) communication enables real-time negotiation between approaching vehicles to safely cross without stopping.

THE CHALLENGE

01. THE HOMOGENEITY ILLUSION

Current autonomous frameworks assume uniform agent behaviors and identical priorities

02. HIGH-STAKES ANOMALIES

The sudden emergence of emergency vehicles (e.g., ambulances) disrupts standard coordination

03. ASYMMETRIC ENVIRONMENTS

Traditional Multi-Agent Reinforcement Learning (MARL) struggles to guarantee instantaneous right-of-way without triggering system-wide gridlock.

THE MATHEMATICAL FRAMEWORK

- Markov Games (Stochastic Games): The environment is modeled as a multi-player sequential optimization framework defined by:

$$\langle S, A_1, \dots, A_n, P, R_1, \dots, R_n \rangle$$

- Joint Action Space: Transition probability depends on the joint action vector $a = (a_1, \dots, a_n)$ of all active agents.
- Non-Stationarity: The environment is non-stationary because other agents alter their policies concurrently as they learn.



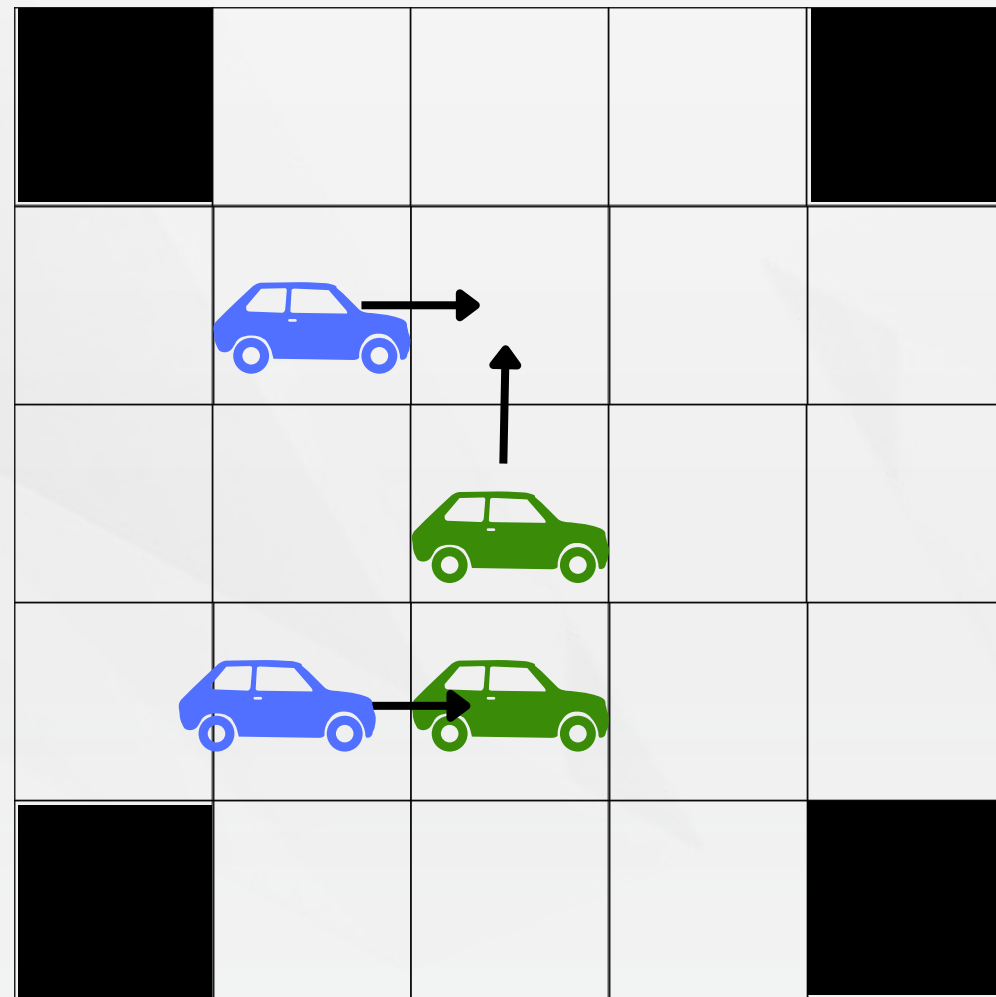
REWARD SHAPING FOR PRIORITY

- Asymmetric Reward Design: Introducing heterogeneous target behaviors into standard MARL.
- Standard Objective: Base rewards maximize forward velocity and heavily penalize spatial overlap (collisions).
- Priority-Aware Penalty: Standard agents receive an acute negative penalty (α) if their actions obstruct the predicted pathway of an active emergency agent:

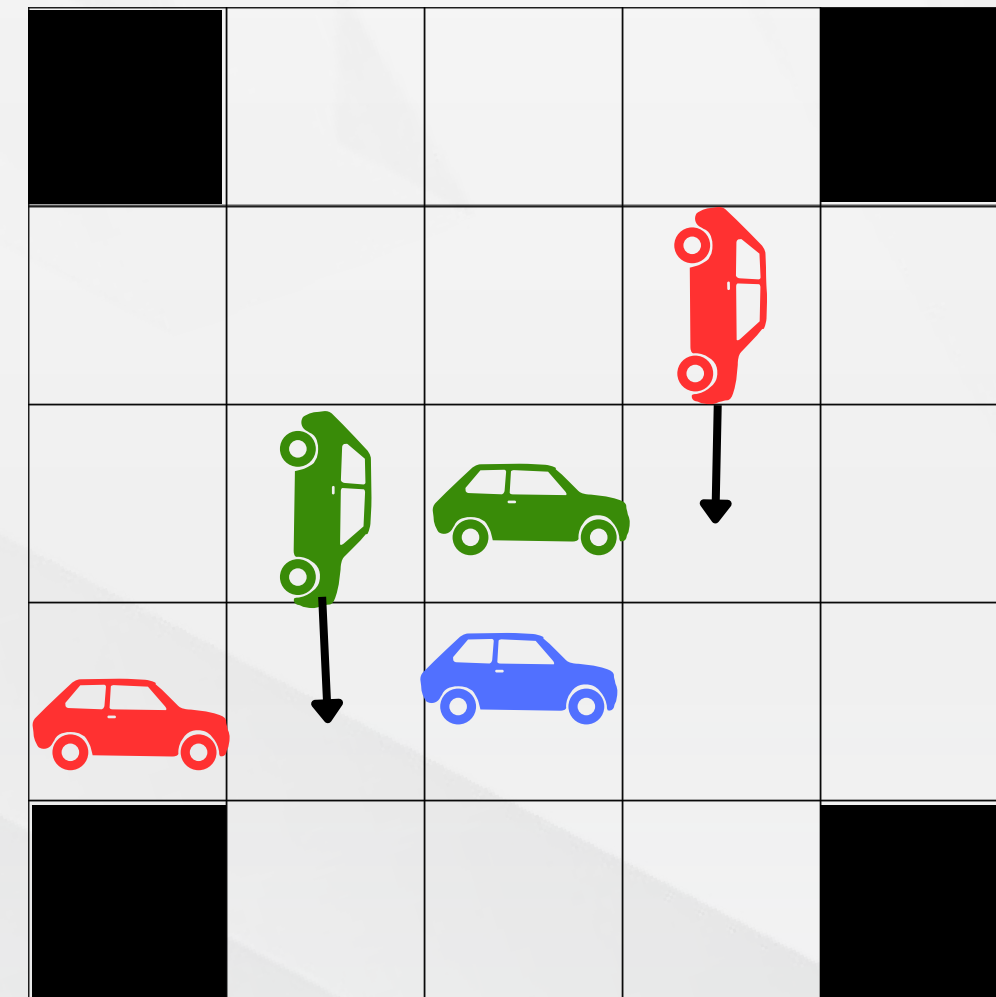
$$R_i(s, a) = R_{speed}(s, a_i) - R_{collision}(a) - \alpha \cdot I(Path\ Obstructed)$$

WORLD DEFINITIONS

- Definition of accident



- Ambulance in first priority



LEARNING DYNAMICS

- **Core Idea:** Each agent adjusts how fast it changes its policy depending on whether it is currently "winning" or "losing" relative to its own average policy.
- **Asymmetric Learning Rates:** When winning, the agent shifts its policy slowly and cautiously, when losing, it shifts quickly to escape a bad strategy.
- The reward structure creates a clear, asymmetric priority order (yield to the ambulance), the environment behaves more like a **coordination game** than an adversarial one.

WOLF-PHC & IQL

Dimension	WoLF-PHC	Independent Q-Learning
Mechanism	Policy-based, variable learning rate ($\delta_{\text{win}} / \delta_{\text{lose}}$)	Value-based, single fixed learning rate (α)
Sensitivity to state-space growth	Higher — avg-policy tracking needs more repeated visits per state	Lower — single Q-update per visit is cheaper to learn from
Computational overhead per step	Slightly higher (extra policy bookkeeping)	Lower

EVALUATION METRICS

SUCCESS RATE

Fraction of episodes where all agents reach their goals with zero collisions.

This is the metric that most honestly reflects real-world safety, since it doesn't average away rare failures the way reward does.

CRASH RATE

Fraction of episodes ending in a collision.

This second case was missing from our initial collision check and silently inflated our early results.

COLLECTIVE REWARD

Sum of all agents' rewards per episode, smoothed over a rolling window.

Useful for tracking overall learning progress, but can hide rare catastrophic failures.

EXPERIMENTAL SCENARIOS

Two Symmetric Vehicles



Two Vehicles + Emergency Vehicle.

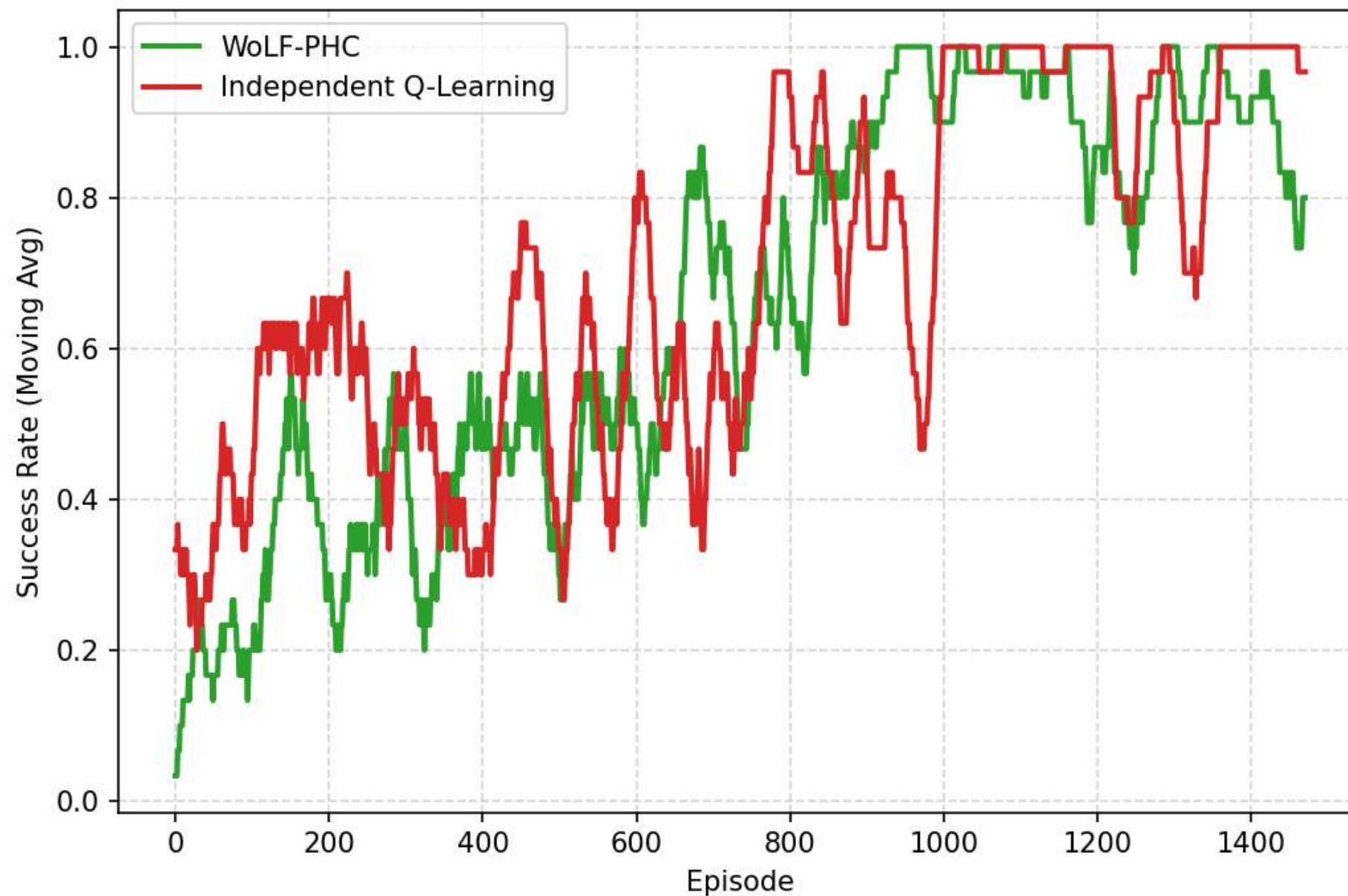


Real life scenarios: multiple
Vehicles + Emergency Vehicle

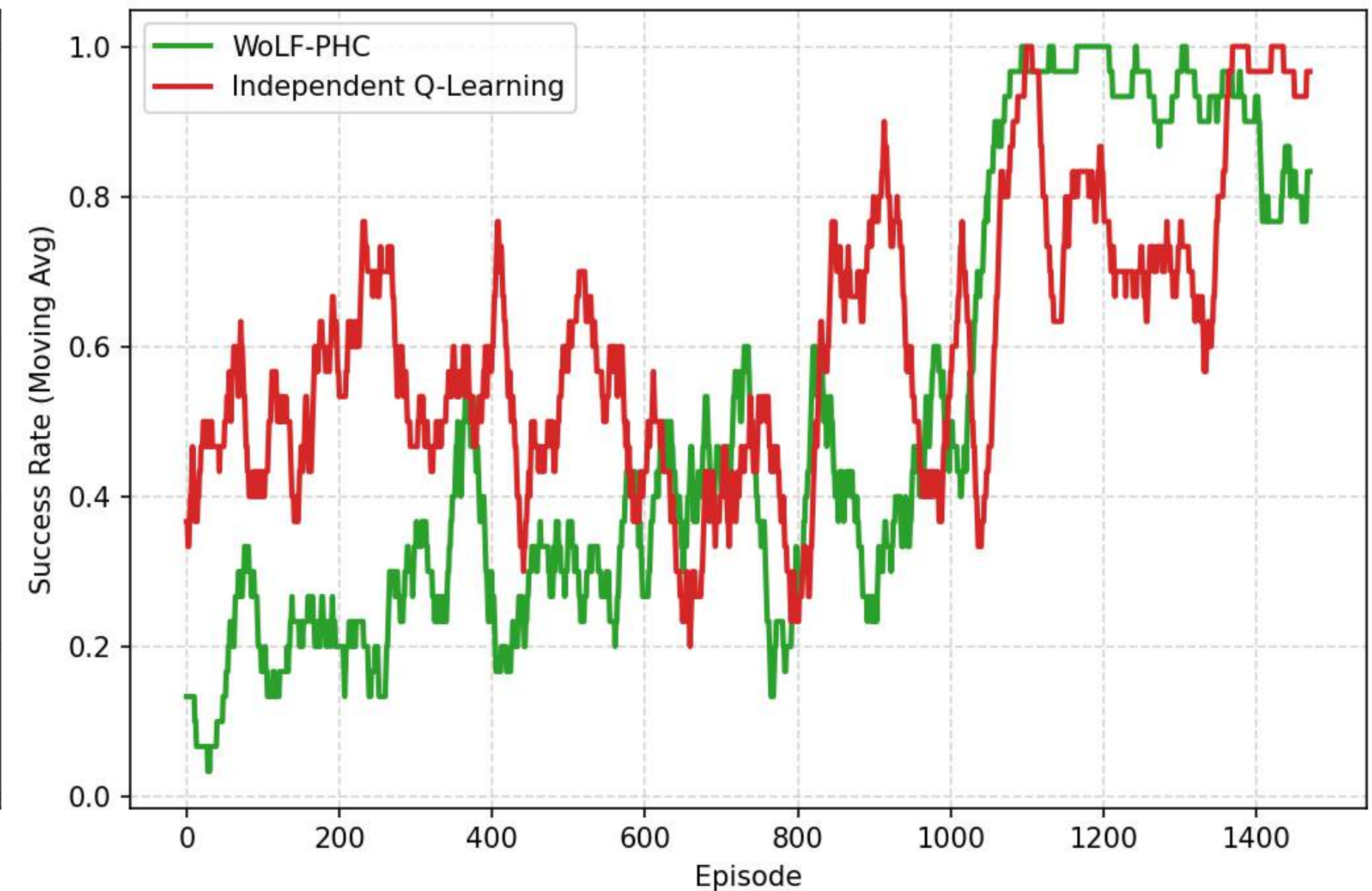
Pure	4	8	12	16
Emergency vehicle	3+1	7+1	11+1	15+1

PRIMARY EXPERIMENTS - RESULTS

Scenario 1: 2 Standard Vehicles (Pure)



Scenario 2: 2 Vehicles + Emergency Vehicle



PRIMARY EXPERIMENTS - FINDINGS

Scenario 1 (2 Standard Vehicles - Pure):

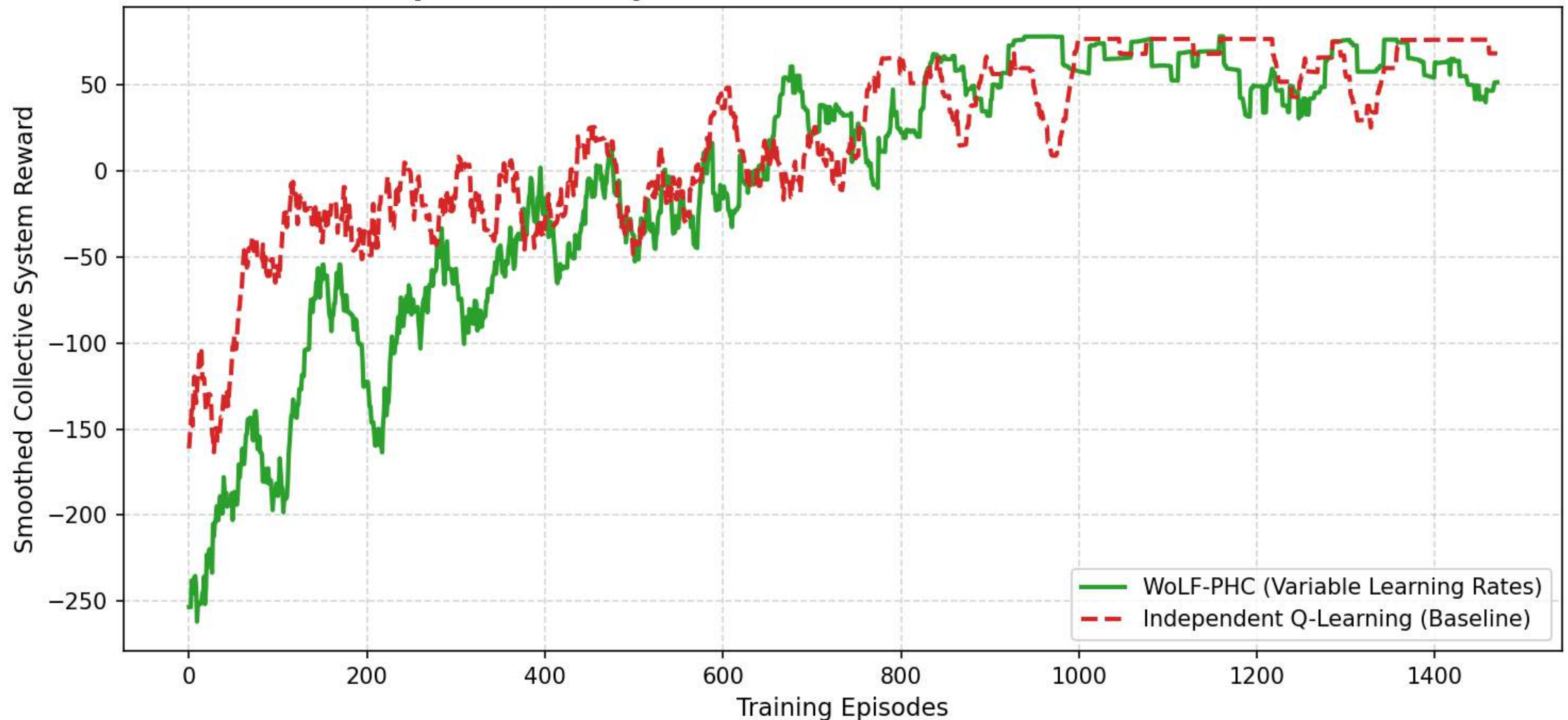
- Both WoLF-PHC and Independent Q-Learning (IQL) successfully learn collision-free navigation policies, converging to a 100% Success Rate (1.0) around Episode 900.
- Stability Contrast: WoLF-PHC demonstrates initial conservatism during early exploration (Episodes 0–400), but once converged, it maintains a stable, high-performing policy with minimal performance degradation.

Scenario 2 (2 Vehicles + Emergency Vehicle):

- WoLF-PHC Convergence: Reaches a stable 100% success rate around Episode 1,100 by successfully learning implicit yielding behaviors for the preemption vehicle.
- IQL Instability: Exhibits higher volatility and periodic drops in success rate due to its inability to adapt learning rates dynamically in non-stationary emergency contexts.

PRIMARY EXPERIMENTS - RESULTS

Comparative Analysis: Scenario 1: 2 Standard Vehicles (Pure)



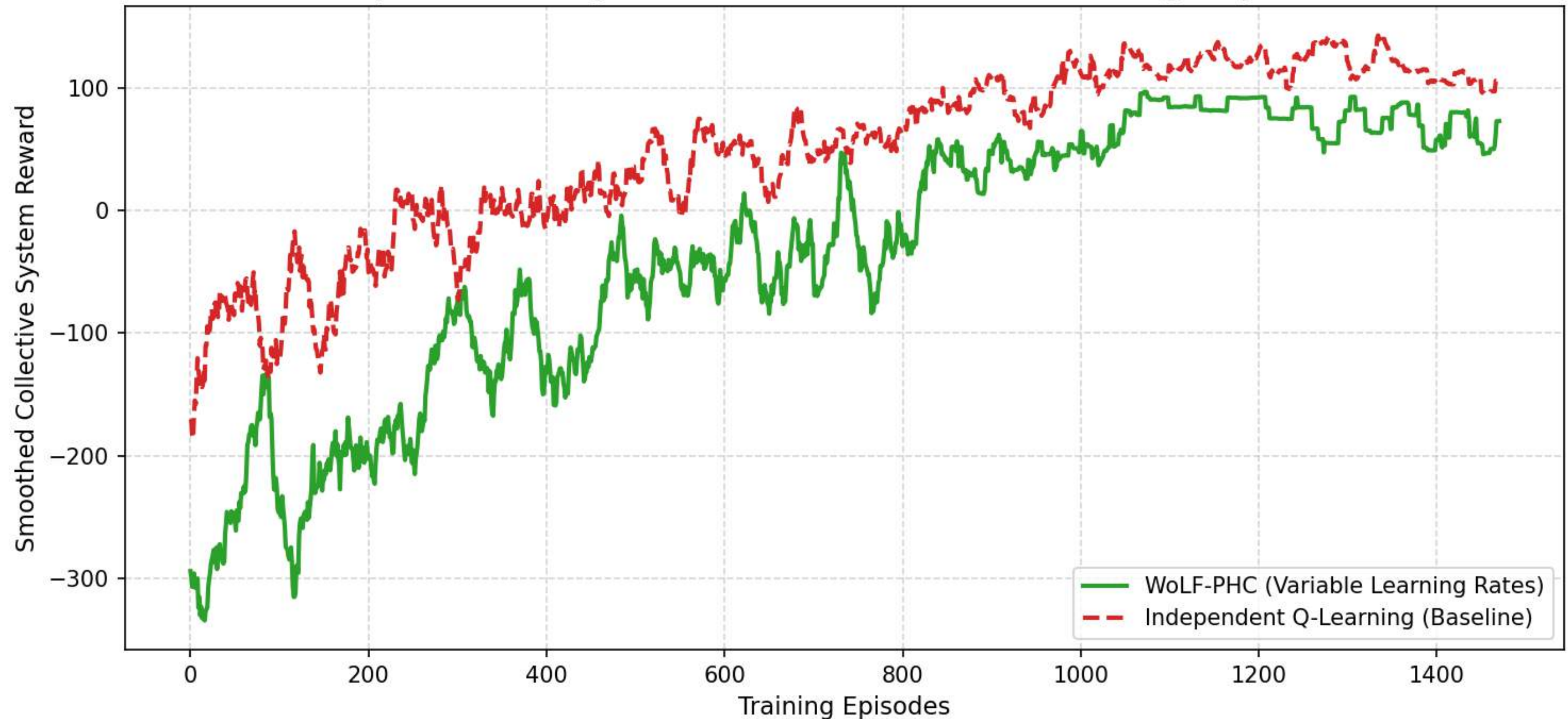
PRIMARY EXPERIMENTS - FINDINGS

SCENARIO 1 (PURE SCENARIO):

- Both algorithms converge to a similar positive reward ceiling (~60–75) past Episode 800.
- IQL demonstrates faster early reward progression (rising from -150 to 0 by Episode 200), whereas WoLF-PHC achieves a smoother, more consistent asymptote in later episodes.

PRIMARY EXPERIMENTS - RESULTS

Comparative Analysis: Scenario 2: 2 Vehicles + Emergency Vehicle



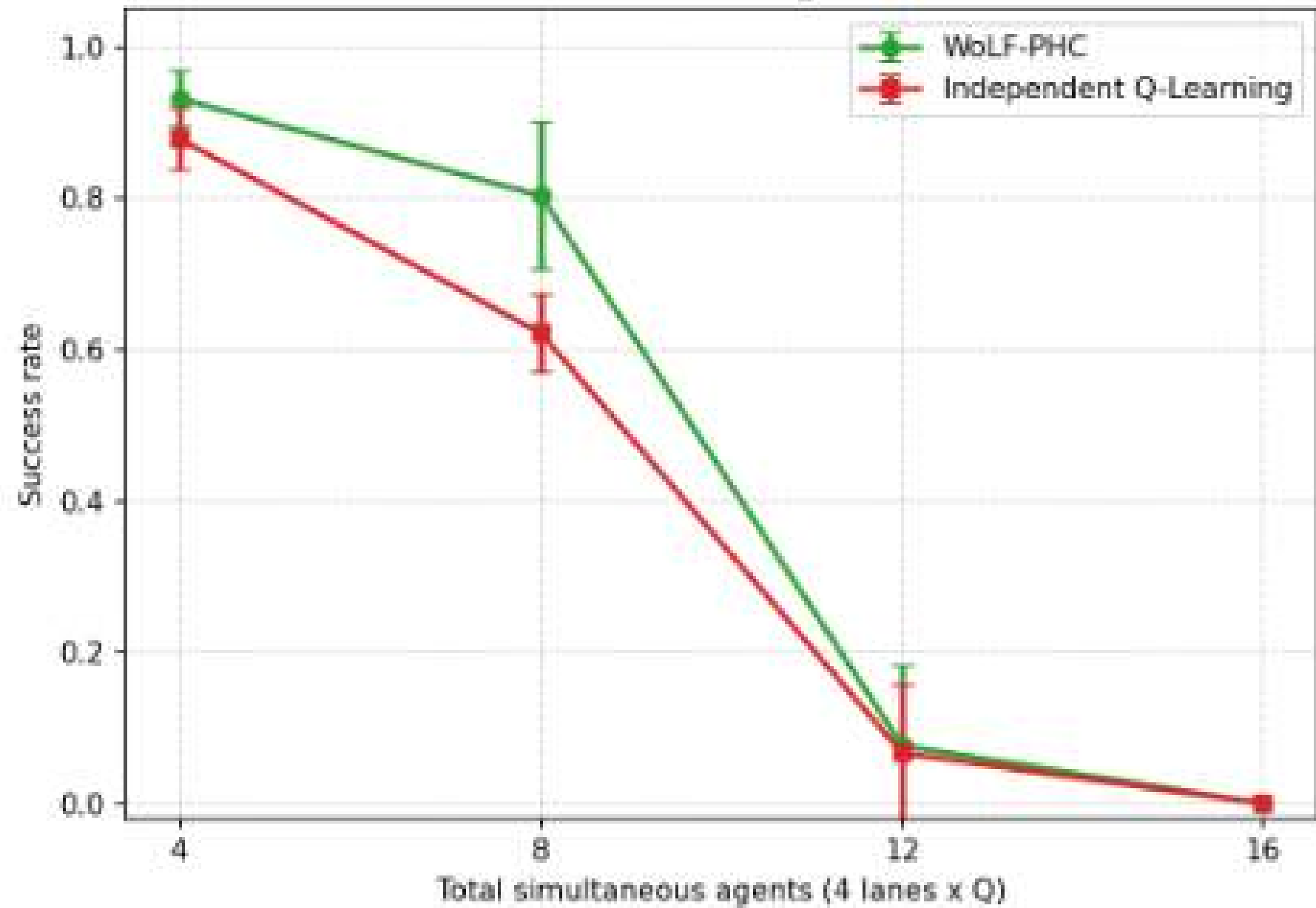
PRIMARY EXPERIMENTS - FINDINGS

SCENARIO 2 (EMERGENCY VEHICLE SCENARIO):

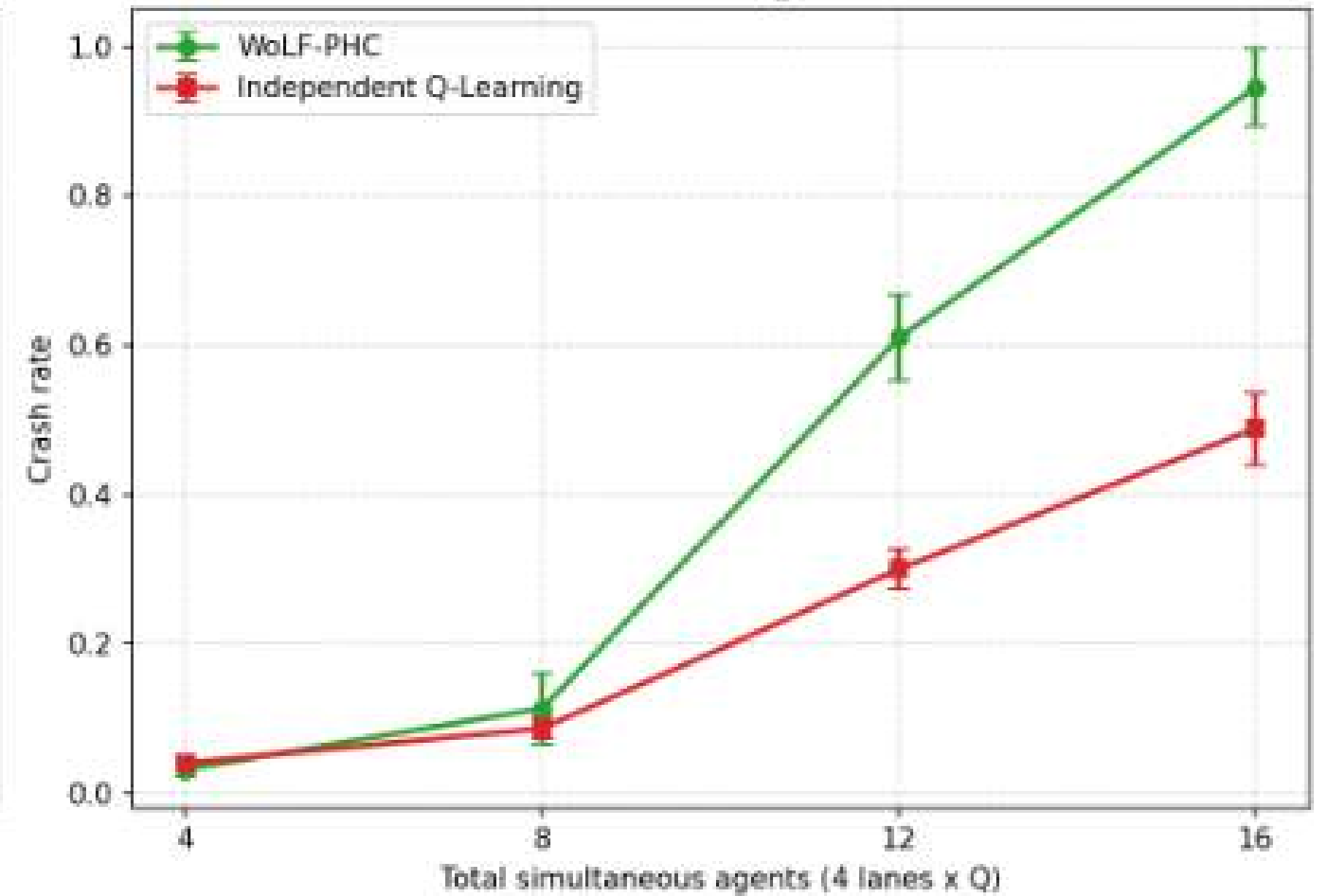
- IQL Nominal Reward Supremacy (~120 vs. ~80): IQL operates egocentrically, attempting to maximize immediate individual rewards without risk-averse adjustments, yielding higher raw reward scores in non-collision trajectories.
- WoLF-PHC Conservative Policy: By dynamically tuning its learning rates, WoLF-PHC prioritizes collision avoidance with the high-priority ambulance over aggressive maneuvers, resulting in a slightly lower nominal reward baseline but significantly higher operational safety.

MULTI AGENT - RESULTS

Pure Scalability (no priority)
Success Rate vs. Agent Count



Pure Scalability (no priority)
Crash Rate vs. Agent Count



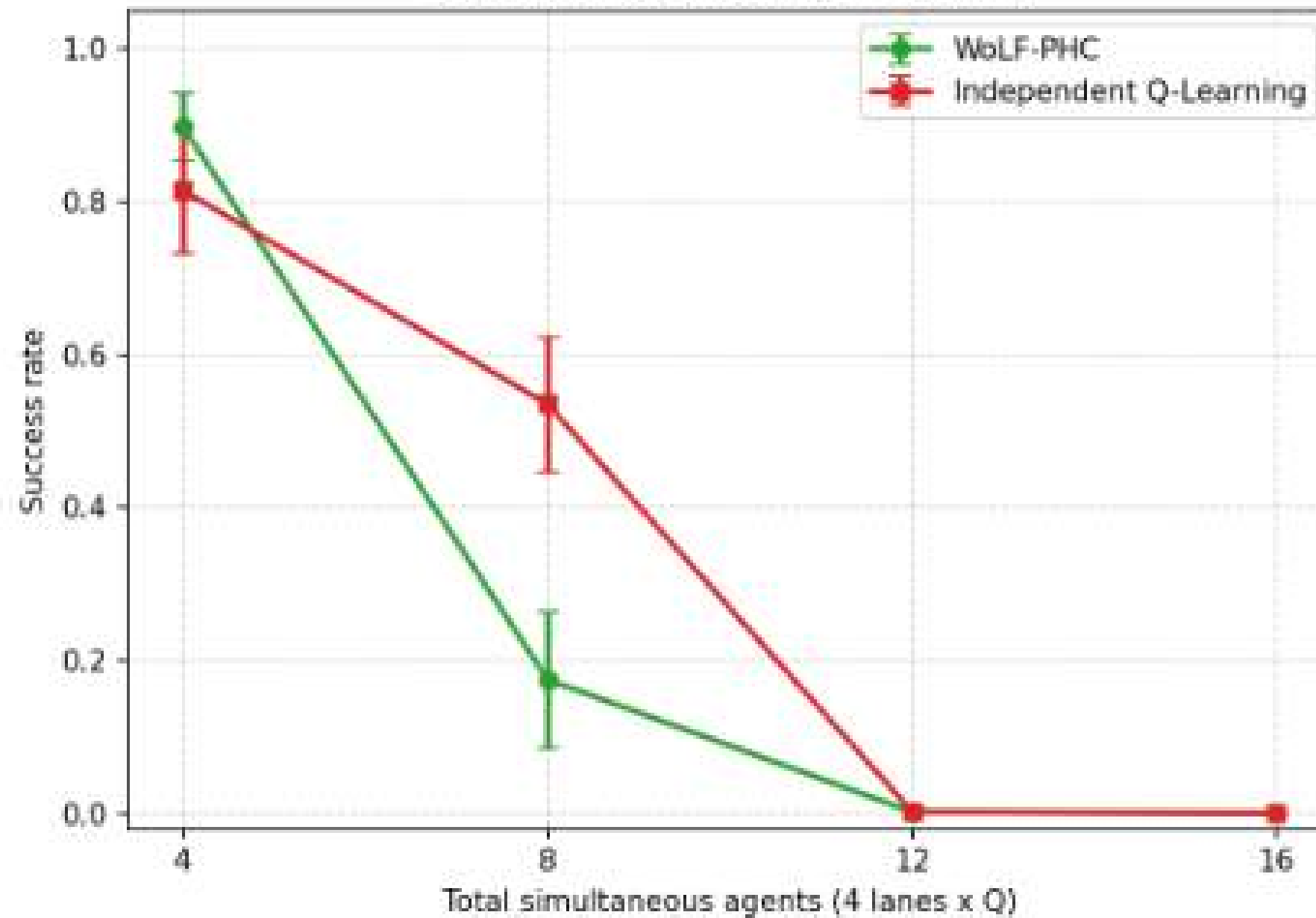
MULTI AGENT - RESULTS

PURE SCALABILITY (NO PRIORITY):

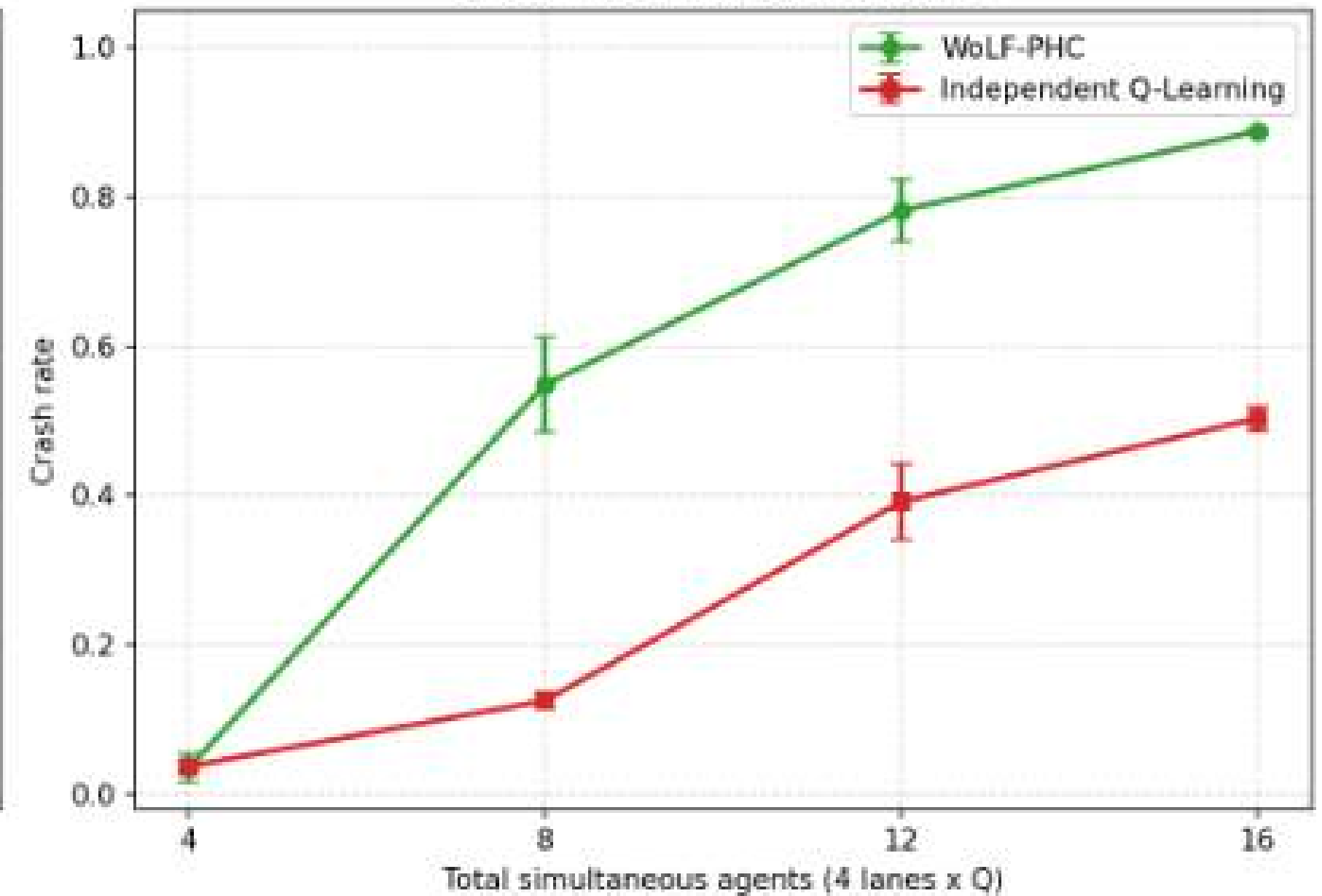
- Small Scale (4–8 Agents): Both algorithms perform well, with WoLF-PHC outperforming Independent Q-Learning (IQL) in Success Rate (~80% vs. ~60% at 8 agents) while keeping Crash Rates low (~10%).
- High Density Crisis (12–16 Agents): As agent density scales up, state-space explosion causes the Success Rate for both models to drop near 0%.
- The WoLF-PHC Paradox: Under extreme congestion (16 agents), WoLF-PHC shows higher Crash Rates (~95%) than IQL (~50%) because its variable learning rate stalls policies in overly cautious gridlocks, causing timeout collisions.

MULTI AGENT - RESULTS

Scalability + Random Ambulance
Success Rate vs. Agent Count



Scalability + Random Ambulance
Crash Rate vs. Agent Count



MULTI AGENT - RESULTS

SCALABILITY + EMERGENCY VEHICLE (RANDOM AMBULANCE):

- Priority Interference: Introducing an ambulance creates additional coordination friction and asymmetry.
- Rapid Performance Decay: WoLF-PHC's Success Rate sharply drops at just 8 agents (~18% vs. ~53% for IQL) with Crash Rates escalating quickly to ~55%.
- IQL's Greed Advantage: Baseline IQL degrades more linearly because simple greedy Q-updates allow agents to clear the intersection individually, whereas complex WoLF policy adaptations fail to converge under dense non-stationarity.

SYNTHETIC DATASET

- the dataset is actively generated via real-time agent interactions (1,500 training episodes per experiment).
- The dataset columns are: Episod, Step, Pos. of Agent 1, Act. of Agent 1, ... , Pos. of Agent n, Act. of Agent n, Reward System, Collision (boolean).
- This dataset' value can be used to imitation learning / behavioural cloning of autonumose vehicels.

REFERENCES

- Shalev-Shwartz, S., Shammah, S., & Shashua, A. (2016). **Safe, multi-agent, reinforcement learning for autonomous driving.** arXiv preprint arXiv:1610.03295.
- Bowling, M., & Veloso, M. (2002). **Rational and convergent learning in stochastic games.** Artificial Intelligence, 136(2), 215-247.
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- Shalev-Shwartz, S., Shammah, S., & Shashua, A. (2017). **On a formal model of safe and scalable self-driving cars.** arXiv preprint arXiv:1708.06374.



THANK YOU